

Flight Delay Prediction



Abstract

Flight delay prediction has become an important research area in the aviation industry due to its significant impact on airline operations, airport efficiency, and passenger satisfaction. Flight delays may lead to increased operational costs, scheduling disruptions, and inefficient resource utilization. Although numerous studies have applied machine learning techniques to predict flight delays, many have focused on specific airports, limited datasets, or individual prediction models, reducing the generalizability of their findings. Therefore, a comprehensive comparison of different machine learning techniques using large-scale flight data remains necessary.

This study aims to predict flight delay status using three supervised machine learning classification models: Logistic Regression, Decision Tree, and Random Forest. These models were selected because they represent different learning approaches, including linear classification, non-linear classification, and ensemble learning. Historical flight records collected in the United States during 2015 were obtained from the Kaggle platform, preprocessed, cleaned, and used for model training and evaluation.

The performance of the models was assessed using several evaluation metrics, including Accuracy, Precision, Recall, F1-Score, ROC-AUC, MCC, Kappa Score, and Log Loss. Experimental results showed that the Random Forest model achieved the highest classification accuracy of 73.8%, outperforming both Logistic Regression and Decision Tree models. The findings indicate that ensemble learning techniques can provide more reliable performance for flight delay prediction tasks. The proposed approach can support airlines in improving operational planning, resource allocation, scheduling decisions, and overall passenger experience.

Keywords: Flight Delay Prediction, Machine Learning, Logistic Regression, Decision Tree, Random Forest.

1. Introduction

Air transportation plays a critical role in modern society by enabling the efficient movement of people and goods across different regions. Despite significant technological advancements in the aviation industry, flight delays remain a major operational challenge that affects airlines, airports, and passengers. Delays can increase operational costs, disrupt airline schedules, reduce airport efficiency, and negatively impact passenger satisfaction. Consequently, accurate flight delay prediction has become an important research topic that supports better planning, resource management, and decision-making within the aviation sector.

In recent years, machine learning techniques have been widely applied to flight delay prediction because of their ability to analyze large volumes of historical flight data and identify complex patterns. Previous studies have demonstrated the effectiveness of machine learning approaches in this area. Choi et al. [2] reported that Random Forest achieved strong performance compared with several traditional classification techniques. Guo et al. [3] proposed a hybrid machine learning framework and demonstrated improvements in prediction accuracy using flight operation data. Afrane et al. [4] showed that Random Forest combined with feature engineering techniques can significantly improve flight delay prediction performance, while Dai et al. [5] demonstrated the effectiveness of hybrid machine learning models for handling complex flight delay datasets.

Although previous studies achieved promising results, several limitations remain. Some studies focused on a single machine learning model without conducting a comprehensive comparison among multiple classification techniques [4], [5]. Other studies relied on specific data sets or limited operational scenarios, reducing the generalizability of their findings [3]. Furthermore, several studies emphasized prediction accuracy while providing limited analysis of other important evaluation metrics such as Precision, Recall, F1-Score, and ROC-AUC [2]. These limitations indicate the need for further investigation and comparison of different machine learning approaches using a large-scale flight delay dataset.

To address these limitations, this study investigates and compares three supervised machine learning classification models: Logistic Regression, Decision Tree, and Random Forest. Logistic Regression was selected as a computationally efficient baseline classifier. Decision Tree was chosen because of its interpretability and ability to capture non-linear relationships within the data. Random Forest was selected because it combines multiple

decision trees to improve prediction accuracy and reduce overfitting through ensemble learning [2], [4].

The dataset used in this study was obtained from the Kaggle platform and contains historical flight records collected in the United States during 2015 [1]. The dataset includes several flight-related attributes, including airline information, departure schedules, arrival schedules, airport information, distance, and delay-related features. Data preprocessing, feature selection, model training, and evaluation were performed before comparing the performance of the selected machine learning models.

The performance of the models was evaluated using Accuracy, Precision, Recall, F1-Score, ROC-AUC, MCC, Kappa Score, and Log Loss. Experimental results showed that Random Forest achieved the highest classification accuracy of 73.8%, outperforming Logistic Regression and Decision Tree models. These findings demonstrate the effectiveness of ensemble learning techniques in addressing flight delay prediction problems and highlight their potential contribution to improving airline operational planning and decision-making processes.

The remainder of this paper is organized as follows. Section 2 presents the related literature review. Section 3 describes the proposed machine learning techniques. Section 4 discusses empirical study, including dataset description and experimental methodology. Section 5 presents experimental results and discussion. Finally, Section 6 concludes the study and provides recommendations for future work.

2. Review of Related Literatures

The prediction of flight delays has attracted much research attention due to its great economic and operational impacts. Delays cost the global aviation industry billions of dollars every year and disrupt millions of passengers. Researchers have applied a wide range of machine learning and statistical techniques to address this challenge, using diverse datasets and evaluation strategies. This section reviews the most relevant published studies, with particular emphasis on work that used the same dataset as this project — the 2015 U.S. domestic flight delays dataset from the Kaggle platform (usdot/flight-delays) — before identifying the key research gaps that motivate the present work.

2.1 Overview of Prior Work

Choi et al. [2] have used Decision Tree, Random Forest and Neural Network models on the U.S. flight and weather data and found that Random Forest had the highest prediction accuracy. However, their analysis only used the accuracy and did not disclose the F1-score and ROC-AUC, which limited the credibility of their evaluation since the class imbalance for the flight delay data was not included. Similarly, Rebollo and Balakrishnan

[12] provide a detailed statistical definition of U.S. air traffic delays, with weather conditions and airport congestion identified as the main causes of delays. However, their work is more oriented towards causal investigation than toward the construction of predictive classifiers.

Guo et al. [3] designed a hybrid Random Forest Regression model using the Maximal Information Coefficient (MIC) feature selection method, and tested it to the data of Beijing International Airport. Their model surpassed Linear Regression, KNN and ANN baselines but was limited to a single airport, raising concerns about its generality. Afrane et al. [4] achieved good results utilizing Random Forest along with improved feature engineering, with an accuracy of 92.36%, F1-score of 91.86%, and ROC-AUC of 96.78%. Despite these impressive results, the study compared only variants of Random Forest rather than evaluating multiple distinct classifiers, which limits its scope as a comprehensive benchmarking study.

Dai et al. [5] proposed a hybrid clustering-based ensemble framework (COWRF with DBSCAN) that reported an average accuracy of 97.2%. While this is state-of-the-art performance, the method is computationally expensive and less interpretable, making it less useful as a practical baseline comparison. Ding [6] used Multiple Linear Regression as the main prediction tool, with about 80% accuracy, but did not attempt non-linear or ensemble based methods. Burgauer and Peters [7] identified distance, day, and departure time as significant predictors in a regression model, though the model was constructed for a single route and showed limited generalizability. Qin and Yu [8] executed a K-means clustering and Fourier modelling at JFK airport. The Fourier model was more accurate, but again was not scalable beyond a single

2.2 Studies Using the Same Dataset

One of the most important studies for this project is the research by AlBassam and AlShahrani [11] that was published in PLOS ONE (December 2025). The authors used exactly the same dataset of U.S. domestic flight delays for 2015 that is used for this project (derived from Kaggle (usdot/flight-delays)). In their study they examined six machine learning classifiers: Decision Tree, Random Forest, Support Vector Classifier (SVC), Logistic Regression, K-Nearest Neighbors (KNN) and Naive Bayes. They focused particularly on the problem of class imbalance. To solve this problem, they utilized three resampling methods including Random Oversampling, SMOTE and ADASYN. Model evaluation was performed using stratified 10-fold cross-validation and multiple metrics including Accuracy, F1-score, Matthews Correlation Coefficient (MCC) and ROC-AUC.

Their results showed that Random Forest combined with Random Oversampling, and Decision Tree combined with SMOTE, both achieved the highest predictive performance with Accuracy of 0.90, F1-score of 0.90, MCC of 0.73, and ROC-AUC of 0.87. These

results represent, to the authors' knowledge, the strongest published performance on this specific dataset to date, and serve as the primary benchmark against which this project's results will be compared in Section 5.

Importantly, the study by AlBassam and AlShahrani did not report the effect of systematic hyperparameter optimization (e.g., GridSearchCV) on the base classifiers prior to resampling, nor did it compare models under a unified tuning framework. This leaves open the question of whether further accuracy gains are achievable through tuning alone, without requiring complex resampling pipelines — which is one of the contributions of the current study.

Table 1: Summary of Related Literature

(★ = uses the same dataset as this study)

Study	Dataset	Models Used	Key Results	Limitations / Gap
Choi et al. (2016)	U.S. flight & weather data	DT, RF, NN	RF achieved highest accuracy	Limited evaluation metrics; no F1/AUC
Rebollo & Balakrishnan (2014)	U.S. air traffic delay data	Statistical models	Identified major delay factors (weather, congestion)	Limited ML implementation
Guo et al. (2021)	Beijing airport data	RF Regression + MIC	Outperformed LR, KNN, ANN	Single airport; no cross-model comparison
Afrane et al. (2025)	Airline & weather data	Random Forest	92.36% accuracy, 91.86% F1, 96.78% AUC	Only RF used; no multi-model comparison
Dai et al. (2024)	Large-scale flight datasets	Hybrid ML (COWRF + DBSCAN)	97.2% accuracy; best hybrid method	Complex structure; high computational cost; low interpretability
Ding (2017)	Airport & weather datasets	Multiple Linear Regression	~80% accuracy	Traditional regression only; no ensemble methods
Burgauer & Peters (2000)	LA–SF airline route data	Multiple Regression	Identified distance, day, departure time as key predictors	Limited to one route; low generalizability
Qin & Yu (2014)	JFK airport data	K-means, Fourier Model	Fourier model achieved better accuracy	Single airport; no scalability

Study	Dataset	Models Used	Key Results	Limitations / Gap
Mueller & Chatterji	Airport delay data	Poisson Distribution	Outperformed ETMS and FACET	Time-dependent accuracy
Rupp (2007)	Airline operational data	Statistical Analysis	Identified operational delay factors	No advanced ML techniques
AlBassam & AlShahrani (2025) ★ Same Dataset	2015 Kaggle U.S. Flight Delays (usdot)	DT, RF, SVC, LR, KNN, Naive Bayes + SMOTE/ADASYN	RF + Random Oversampling: Acc=0.90, F1=0.90, MCC=0.73, AUC=0.87	No hyperparameter tuning reported; no MCC/Kappa in baseline comparison

2.3 Research Gap Analysis

The reviewed literature reveals several recurring limitations. First, many earlier studies evaluated their models using accuracy alone, which is an unreliable metric when the target classes are imbalanced — a well-known characteristic of flight delay datasets where non-delayed flights significantly outnumber delayed ones [1][6][7]. Second, some studies limited their works to one airport, airline route or time window which did not lead to generalizable conclusions [3][7][8]. Third, the majority of studies either applied only one model type or did not perform systematic hyperparameter optimization, making it difficult to assess the true ceiling of model performance [4][5]. With respect to the specific dataset used in this project — the 2015 Kaggle U.S. flight delays dataset — only AlBassam and AlShahrani [11] have published a rigorous multi-classifier evaluation. While their study sets a strong benchmark (RF accuracy 0.90, AUC 0.87 with resampling), it does not investigate the impact of hyperparameter tuning independently of resampling. The present project addresses this gap by applying GridSearchCV-based tuning to the Random Forest classifier on the same dataset, reporting an extended set of metrics including MCC, Kappa score, and Log Loss, and providing a direct comparison with the benchmark results of AlBassam and AlShahrani [11].

Table 2: Research Gap Checklist

Research Gap	AlBassam & AlShahrani (2025)	Choi et al. (2016)	Afrane et al. (2025)	Dai et al. (2024)	This Study
Multiple ML Model Comparison	✓	✓	✗	Limited	✓
Hyperparameter Tuning (GridSearchCV)	✗	✗	✗	Limited	✓
ROC-AUC / F1-Score Evaluation	✓	✗	✓	Limited	✓
MCC & Kappa Score Evaluation	✓	✗	✗	✗	✓
Handling Class Imbalance	✓	✗	Limited	✗	✓
Same 2015 Kaggle Dataset	✓	✗	✗	✗	✓
Large-Scale Generalization (5.8M records)	✓	Limited	Limited	✓	✓

4.0 Description of the Proposed Techniques

This study proposes and compares three supervised machine learning classification techniques to predict whether a flight will be delayed or arrive on time: Logistic Regression, Decision Tree, and Random Forest. These techniques were deliberately selected because they represent three distinct learning paradigms — linear classification, non-linear rule-based classification, and ensemble learning — which allows the problem to be evaluated from complementary modeling perspectives rather than relying on a single approach. All three models were trained on the same set of pre-flight and scheduling-related predictors (described in Section 5) and were configured with balanced class weights to account for the moderate class imbalance in the dataset. The following subsections describe each technique, its underlying principle, the reason for its selection, and the main configuration used in this study, and refer the reader to additional sources for further detail.

4.1 Technique #1: Logistic Regression

Logistic Regression is a linear classification algorithm that estimates the probability that an instance belongs to a particular class by applying the logistic (sigmoid) function to a weighted linear combination of the input features. Given a feature vector x , the model first computes a linear score z and then maps it to a probability using the sigmoid function:

$$z = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n$$
$$P(y = 1 \mid x) = \frac{1}{1 + e^{-z}}$$

A decision threshold (commonly 0.5) is then applied to assign the final class label — in this study, delayed or not delayed. Logistic Regression was selected as the baseline model because it is simple, computationally efficient, and highly interpretable, since the learned coefficients (β) indicate the direction and relative influence of each feature on the prediction. In this work, the model was configured with balanced class weights to reduce bias toward the majority class and with a sufficiently high iteration limit to ensure convergence. A more detailed treatment of Logistic Regression and its application to flight delay prediction can be found in [11].

4.2 Technique #2: Decision Tree

A Decision Tree is a non-linear, rule-based classification technique that recursively partitions the feature space into smaller and more homogeneous regions. At each internal node, the algorithm selects the feature and threshold that best separate the classes according to an impurity measure such as the Gini index, defined for a node t as:

$$Gini(t) = 1 - \sum_{i=1}^C p_i^2$$

where p_i is the proportion of samples belonging to class i at node t , and C is the number of classes. The split that produces the largest reduction in impurity is chosen, and this process continues until a stopping condition is reached, producing leaf nodes that correspond to the final class predictions. The Decision Tree was selected because it is easy to interpret and visualize, requires minimal data preprocessing, and can capture non-linear relationships and feature interactions that a linear model such as Logistic Regression cannot represent. Its main limitation is that a single, fully grown tree tends to overfit the training data, which motivates the ensemble approach presented in the next technique. Further details on Decision Trees and their use in airline delay prediction are provided in [2].

4.3 Technique #3: Random Forest

Random Forest is an ensemble learning technique that constructs a large number of decision trees and combines their outputs to produce a single, more robust prediction. Each tree is trained on a different bootstrap sample of the training data, and at each split only a random subset of features is considered. For a classification task with B trees, the final prediction is obtained by majority voting:

$$\hat{y} = \text{mode} \{ T_1(x), T_2(x), \dots, T_B(x) \}$$

where $T_b(x)$ is the prediction of the b -th tree. This combination of bagging and feature randomness reduces the variance and overfitting associated with individual decision trees while preserving their ability to model non-linear relationships, which generally results in higher and more stable predictive performance. Random Forest was selected as the primary model in this study because of these properties and because prior work has repeatedly identified it as one of the strongest classifiers for flight delay prediction [2], [4]. In this study, the model was configured using 300 trees, a maximum tree depth of 15, a minimum of 10 samples required to split a node, and balanced class weights; these hyperparameters were subsequently refined through the optimization process described in Section 5.4. A more comprehensive description of Random Forest and its application to this domain can be found in [4] and [11].

Comparison of the Three Proposed Techniques

Technique	Learning Paradigm	Key Idea	Main Strengths	Main Limitations	Role in This Study
Logistic Regression	Linear / probabilistic	Models class probability using the sigmoid of a linear combination of features	Simple, fast, interpretable coefficients	Assumes a linear decision boundary; weak with complex interactions	Baseline model
Decision Tree	Non-linear / rule-based	Recursively splits the feature space using an impurity measure (Gini)	Interpretable, captures non-linearity, minimal preprocessing	A single tree tends to overfit	Non-linear comparison model
Random Forest	Ensemble (bagging)	Aggregates many decision trees through majority voting	High accuracy, reduces overfitting and variance, robust	Less interpretable, higher computational cost	Primary (proposed) model

5.0 Empirical Studies

5.1 Description of dataset

The dataset used in this project is from the Kaggle platform. It contains historical airline flight records collected in the United States during the year 2015 [1]. The dataset attributes are year, month, day, day of week, airline, flight number, tail number, origin airport, destination airport, scheduled departure, departure time, departure delay, taxi out, wheels off, scheduled time, elapsed time, air time, distance, wheels on, taxi in, scheduled arrival, arrival time, arrival delay, diverted, cancelled, cancellation reason, air system delay, security delay, airline delay, late aircraft delay, and weather delay.

The target variable in this study is arrival delay, which determines whether a flight is delayed or on time. Several features were selected as predictors, including month, day, day of week, airline, distance, origin airport, destination airport, scheduled departure, and departure time.

The dataset consists of approximately 5.8 million flight records, which is good for a flight delay prediction classification machine learning model. A random sampling technique was applied to reduce the dataset size to 20,000 samples. After sampling, the class distribution was analyzed to ensure that the target variable remained balanced as before sampling. The results showed a slight imbalance, with approximately 64% non-delayed flights and 36%

delayed flights. This imbalance was considered during model evaluation using appropriate performance metrics such as F1-score.

The first step before training the machine learning models is data preprocessing to ensure the best performance of the models; it includes removing rows with missing target, dropping unnecessary columns, filling necessary ones missing values with the median as needed, and encoding categorical variables.

The dataset was divided into training and testing subsets using an 80:20 ratio, where 80% of the records were used for training the models and 20% were used for testing and evaluation.

Table 3: Description of Important Dataset Attributes

Description	Attribute
Airline company operating the flight	AIRLINE
Unique identifier of the flight	FLIGHT_NUMBER
Departure airport	ORIGIN_AIRPORT
Arrival airport	DESTINATION_AIRPORT
Planned departure time	SCHEDULED_DEPARTURE
Delay in departure time (minutes)	DEPARTURE_DELAY
Actual flight duration	AIR_TIME
Distance between airports	DISTANCE
Delay in arrival time (minutes)	ARRIVAL_DELAY
Delay caused by weather conditions	WEATHER_DELAY
Delay caused by airline operations	AIRLINE_DELAY

5.1.1 Statistical Analysis of the Dataset

To understand the distribution of the dataset statistical analysis was performed, so it helps identify important patterns related to flight delays. Measures such as mean, median, standard deviation, minimum, and maximum values were computed for the main numerical attributes.

The analysis showed that the average departure delay was approximately 9.37 minutes, while the average arrival delay was approximately 4.41 minutes. The large standard deviation values indicate high variability in flight delays, which confirms the complexity of predicting delays.

In addition, the median values of both departure delay (-2 minutes) and arrival delay (-5 minutes) indicate that many flights arrived slightly earlier than scheduled, while a smaller number of flights experienced very large delays.

Table 4: Statistical Analysis of Selected Features

	Mean	Median	Standard deviation	Maximum	Minimum
Departure Delay	9.37	-2	37.08	1988	-82
Arrival Delay	4.41	-5	39.27	1971	-87
Air Time	113.51	94	72.07	690	7
Distance	821.00	647	609.40	4983	21

5.1.2 Correlation Analysis

The purpose of correlation analysis is to identify relationships between the selected features and flight delays. The results indicate that departure features such as departure delay and other delay-related variables have a very strong positive correlation with arrival delay, which indicates a potential data leakage issue. To prevent target leakage, features directly derived from arrival delay or post-flight information were excluded from the model input features. Only pre-flight and scheduling-related variables were retained for prediction.

Table 5: Correlation Between Features and Arrival Delay

Correlation with Arrival Delay	Feature
0.944	Departure Delay
0.609	Airline Delay
0.522	Late Aircraft Delay
0.265	Weather Delay
0.247	Air System Delay
-0.025	Distance

5.1.3 Dataset Challenges

Preprocessing and analyzing a large and comprehensive dataset can lead to several challenges [1], including:

Missing values in some delay-related attributes.

Imbalanced distribution between delayed and on-time flights.

These challenges required preprocessing and cleaning techniques before training the machine learning models to get correct, useful results.

5.2. Experimental Setup

The experiment was carried out using the Python programming language in Visual Studio Code. Several libraries were used in this project, including Pandas and NumPy for data handling, Matplotlib and Seaborn for visualization, and Scikit-learn for machine learning model training and evaluation.

The flight delay dataset was first loaded into the system and then inspected to understand the structure of the data, including column names, data types, and missing values. After the data inspection process, the target variable was created by classifying flights into two categories: Flights with an arrival delay greater than zero were labeled as delayed, while the remaining flights were labeled as not delayed.

Three supervised machine learning classification models were implemented in this experiment: Logistic Regression, Decision Tree, and Random Forest.

Logistic Regression was used as a baseline classification model. Decision Tree was selected because it is easy to interpret and visualize. Random Forest was selected because it combines multiple decision trees to improve prediction performance and reduce overfitting.

The models were trained using the training dataset and evaluated using the testing dataset.

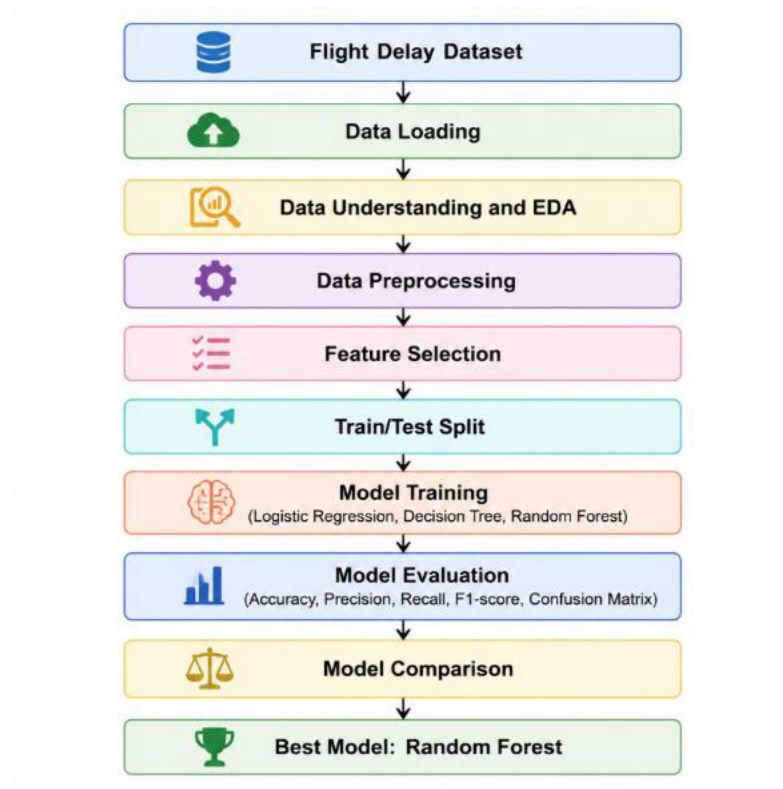


Figure 1: Methodology diagram.

Table 6: Experimental Setup

Item	Description
Programming Environment	Visual Studio Code
Programming Language	Python
Dataset	Flight Delay Dataset
Sample Size	20,000
Models	Logistic Regression, Decision Tree, Random Forest
Train/Test Split	80% / 20%
Evaluation Metrics	Accuracy, Precision, Recall, F1-score, Confusion Matrix

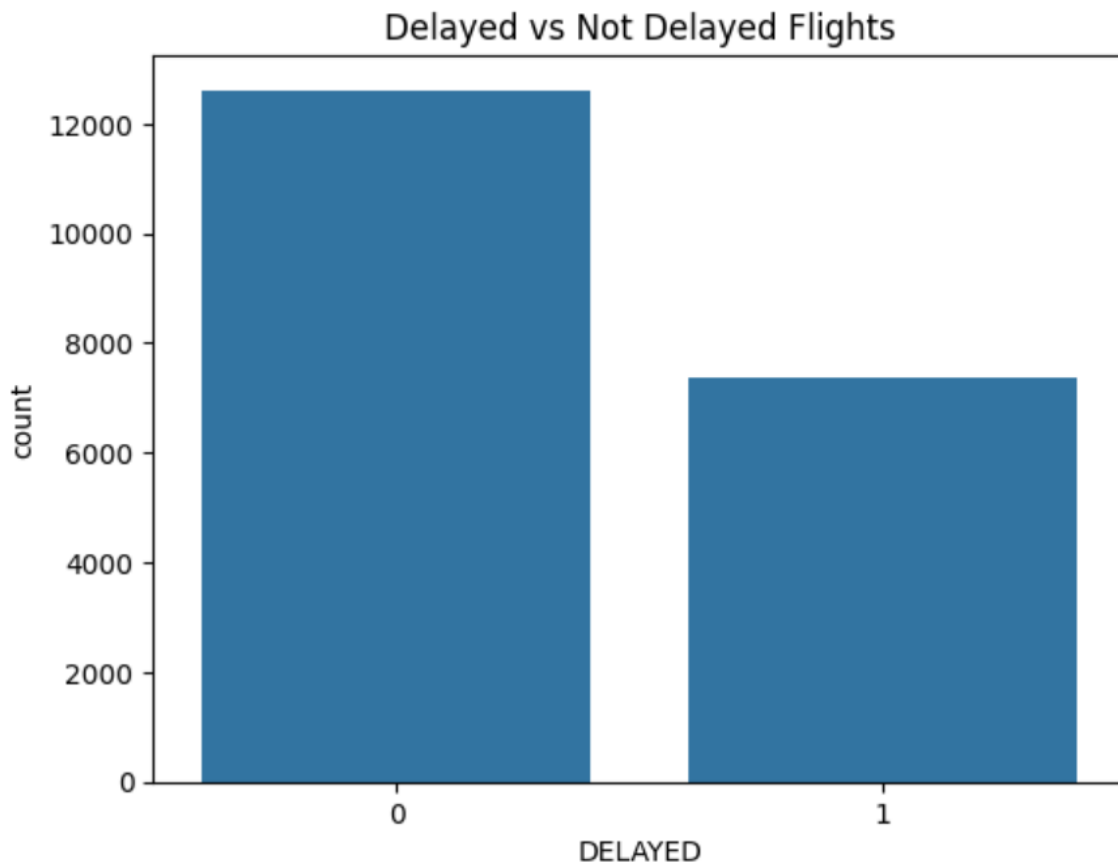


Figure 2: *Distribution of delayed and non-delayed flights.*

5.3 Performance Measures

The three classification models were evaluated using several common measures to make sure the results are reliable. These measures are widely used in studies on predicting flight delays. The accuracy of the models shows how many flights were correctly classified out of all the predictions. This gives an idea of how well the models work, but it can be misleading when there are more non-delayed flights than delayed ones, where there are a lot more non-delayed flights. The precision of the models shows how many flights that were predicted to be delayed were actually delayed. A higher precision means there are fewer false alarms, which is important for airlines that make decisions based on delay predictions. The recall of the models shows how many actual delayed flights were correctly identified. This measure is very important because if a real delay is missed, it can affect passengers and airline schedules.

The F1-score is a measure that takes into account both precision and recall. It is especially useful for projects where there are more non-delayed flights than delayed ones, as in this case, as shown in other studies.

The ROC-AUC or Receiver Operating Characteristic, Area Under the Curve, measures how well the models can tell the difference between delayed and nondelayed flights. A value closer to 1.0 means the model is better at telling the difference.

In addition, the confusion matrix was used to get a breakdown of the true positives, true negatives, false positives, and false negatives for each model.

Table 7: Model Performance Comparison.

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC	MCC	Kappa	Log Loss
Logistic Regression	0.6670	0.5253	0.6231	0.5700	0.7196	0.3045	0.3015	0.6485
Decision Tree	0.7133	0.5898	0.6260	0.6073	0.6936	0.3822	0.3818	10.3355
Random Forest	0.7382	0.6670	0.5215	0.5853	0.7574	0.4047	0.3983	0.5719

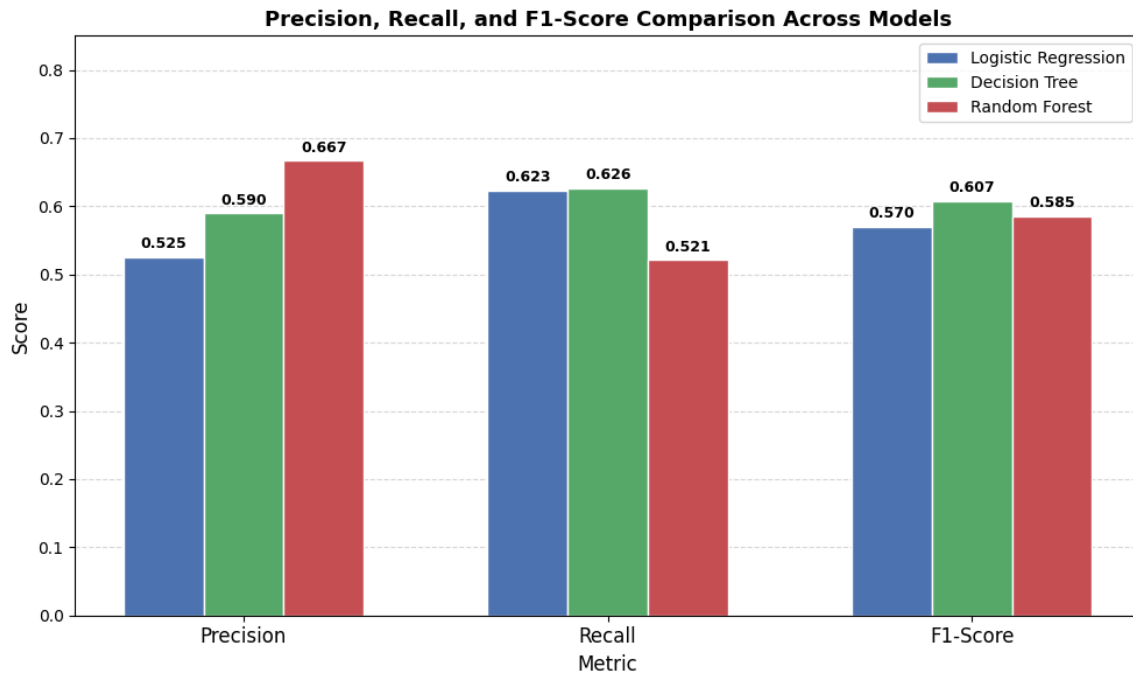


Figure 3: Precision, recall, and F1-cscore for all three classification models.

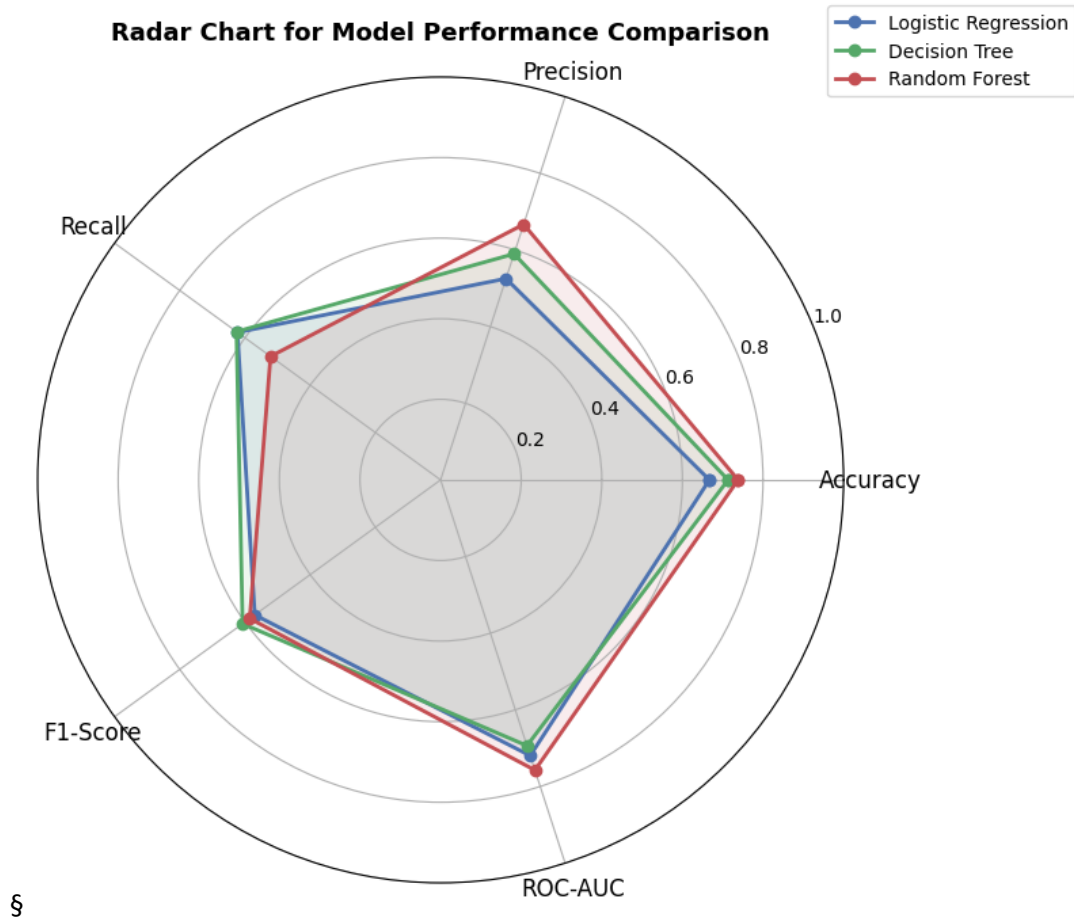


Figure 4: Radar Chart for Model Performance Comparison.

The results show that Random Forest achieved the best overall performance among all tested models. The better performance of it can be explained by its ensemble learning approach, where multiple decision trees are combined to improve prediction accuracy and reduce overfitting.

Table 8: Best-Performing Model for Each Evaluation Metric.

Evaluation Metric	Best Score	Best Model
Accuracy	0.7382	Random Forest
Precision	0.6670	Random Forest
Recall	0.6260	Decision Tree
F1-Score	0.6073	Decision Tree
ROC-AUC	0.7574	Random Forest
MCC	0.4047	Random Forest
Kappa	0.3983	Random Forest
Log Loss	0.5719 (lower is better)	Random Forest

The bar chart shows that it obtained the highest accuracy compared to Logistic Regression and Decision Trees.

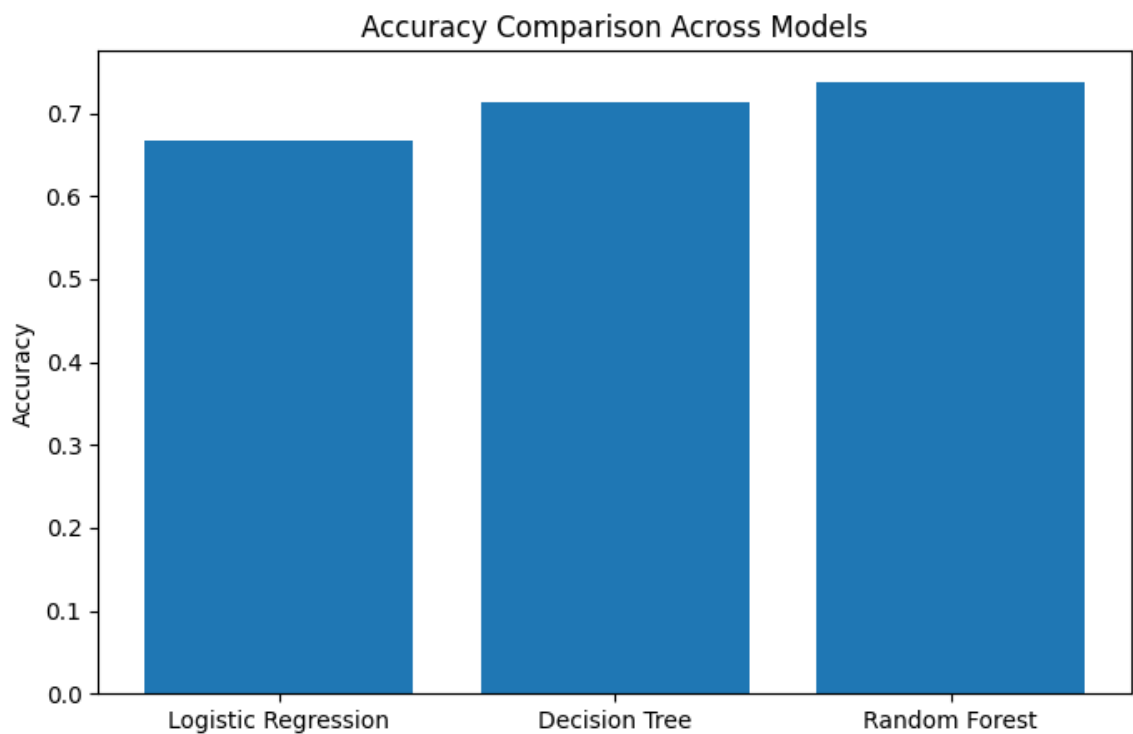


Figure 5: Accuracy for all three classification models.

The confusion matrix shows that the Logistic Regression model correctly classified 1785 non-delayed flights and 883 delayed flights; the Decision Tree model correctly classified 1966 non-delayed flights and 887 delayed flights; and the Random Forest model correctly classified 2214 non-delayed flights and 739 delayed flights. However, some flights were still incorrectly classified, resulting in false positives and false negatives.

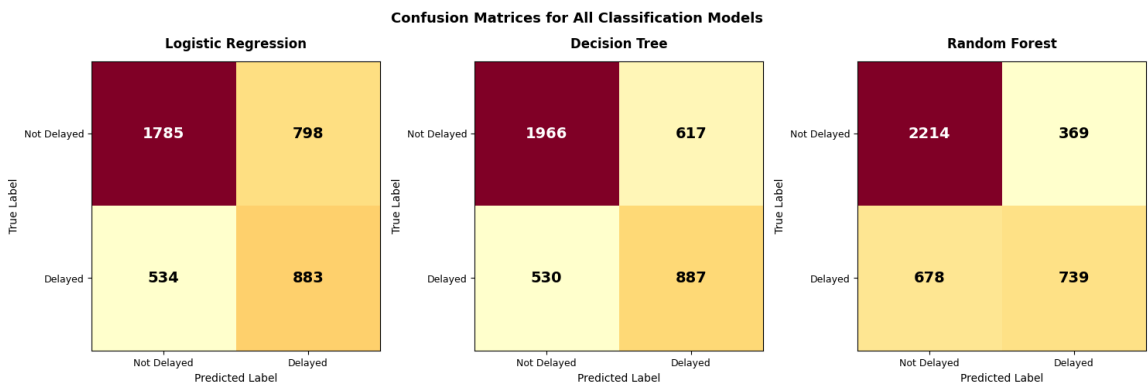


Figure 6: Confusion Matrices for all three classification models.

The Random Forest model was the best at telling the difference with a ROC-AUC score of 0.7574, which is better than the Logistic Regression model 0.7196 and the Decision Tree model 0.6936.

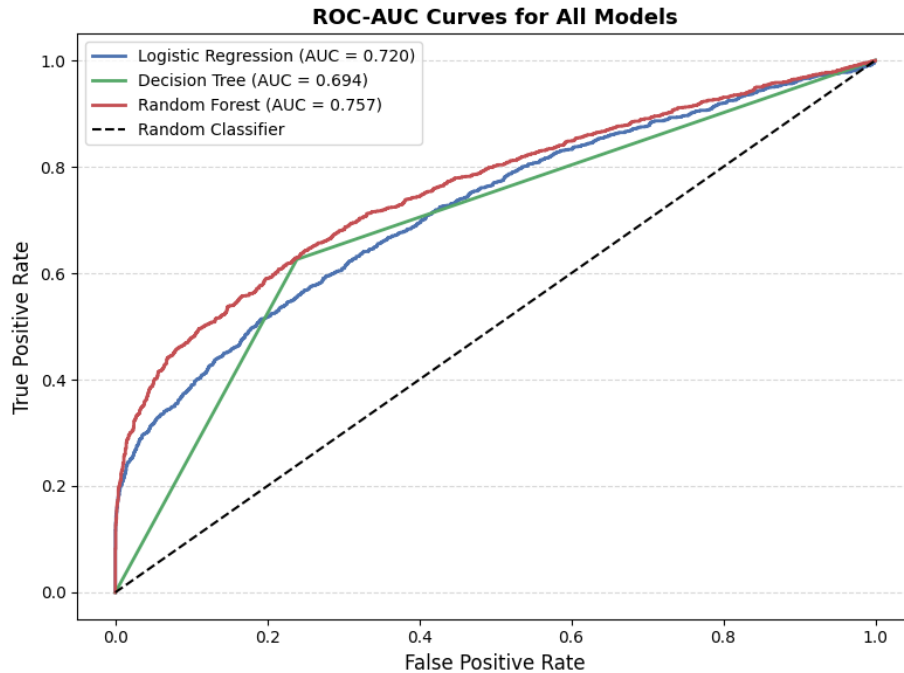


Figure 7: ROC-AUC curves for all three classification models.

5.4. Optimization strategy

To improve the performance of the Random Forest classifier, hyperparameter tuning using Grid Search with 5-fold cross-validation was used to identify the optimal combination of parameters that maximizes classification accuracy.

The parameters evaluated during the search are the number of trees in the forest (`n_estimators`), the maximum depth of the trees (`max_depth`), and the minimum number of samples required to split a node (`min_samples_split`).

The results showed that increasing the number of estimators improved model performance up to a certain point, while controlling tree depth helped balance model complexity and reduce overfitting and select an appropriate minimum number of samples for splitting, improving generalization.

The best-performing configuration achieved a cross-validation accuracy of 0.7487, demonstrating a stronger predictive performance than baseline models.

Table 9: Optimum parameters for Random Forest Model model.

Parameter	Optimal Value
n_estimators	400
max_depth	20
min_samples_split	10
random_state	42
class_weight	balanced

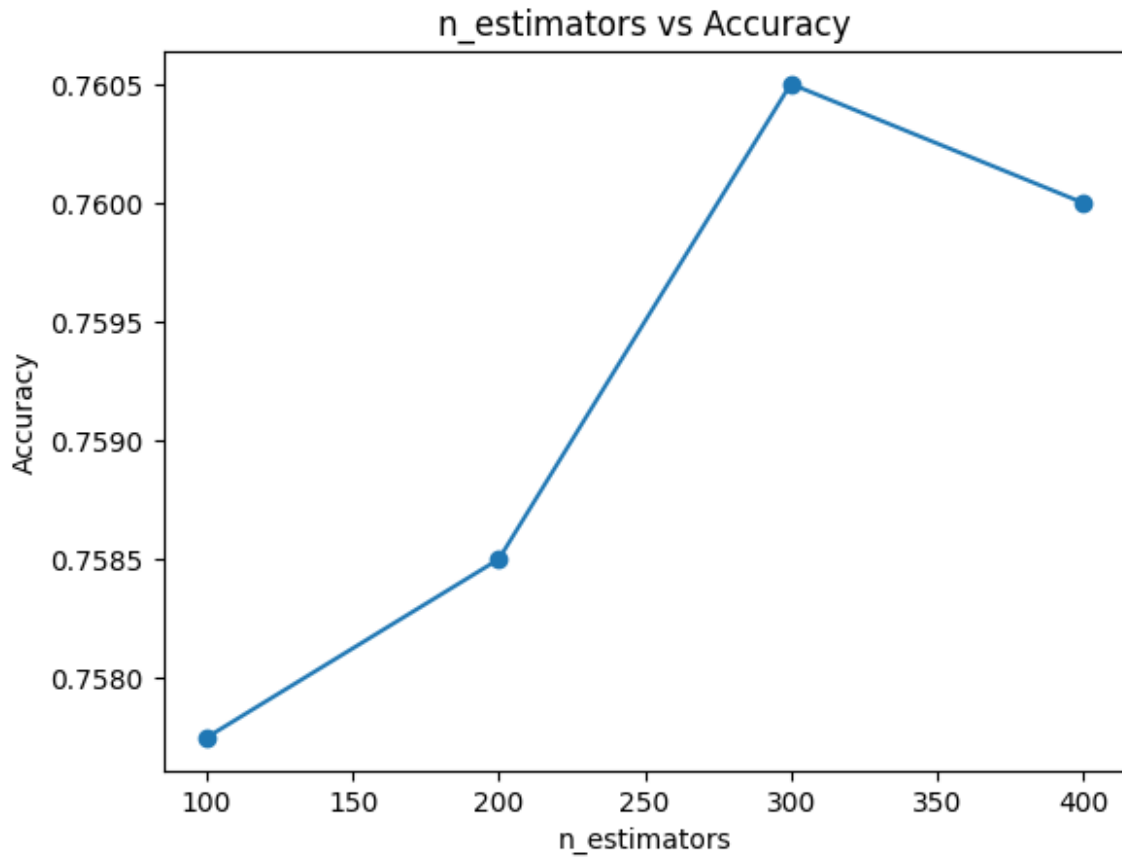


Figure 8. Random Forest Hyperparameter Search for Number of Trees (n_estimators).

This figure indicates that a few trees result in low accuracy. and it improves as the number of trees increases until approximately 300 trees, suggesting that is the optimal balance between performance and computational cost.

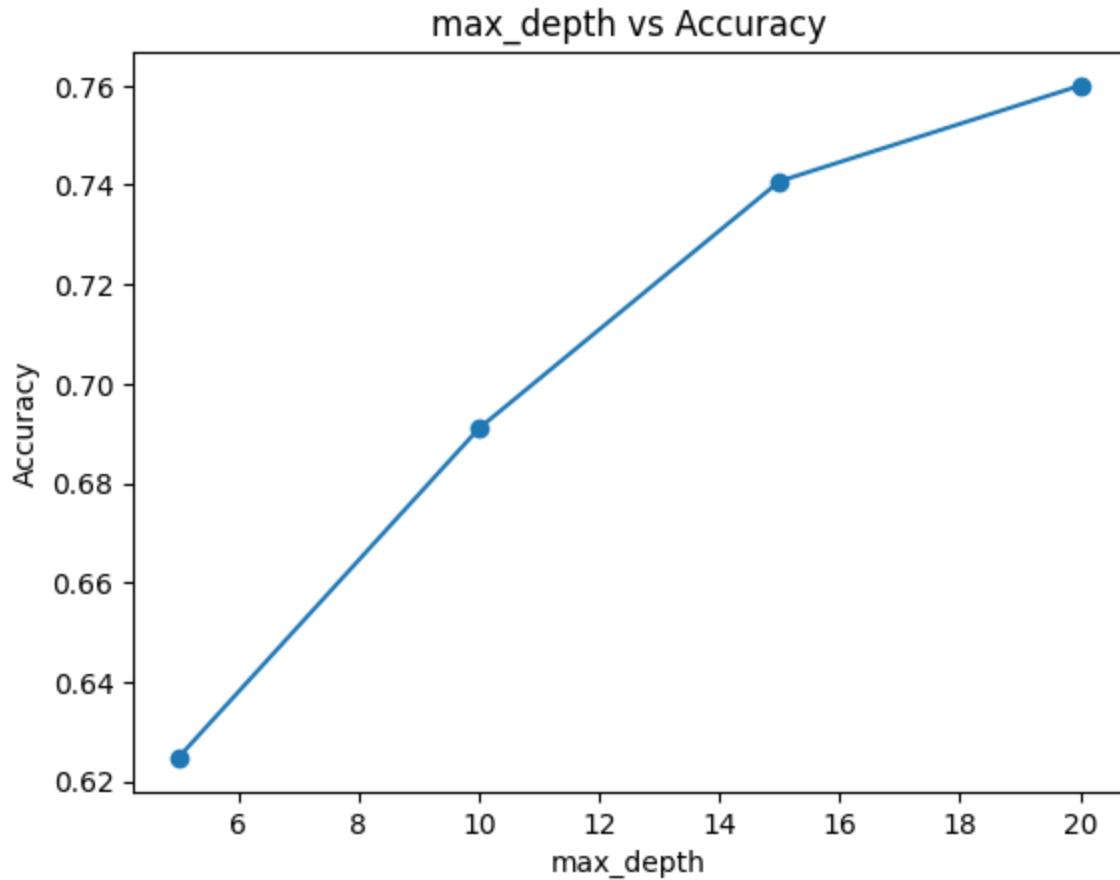


Figure 9. Random Forest Hyperparameter Search for Maximum Tree Depth (max_depth) .

This figure indicates that shallow trees result in underfitting and lower performance, while increasing the depth improves accuracy up to an optimal point. Beyond this point, deeper trees may lead to overfitting, reducing generalization performance. The optimal depth provides the best trade-off between bias and variance.

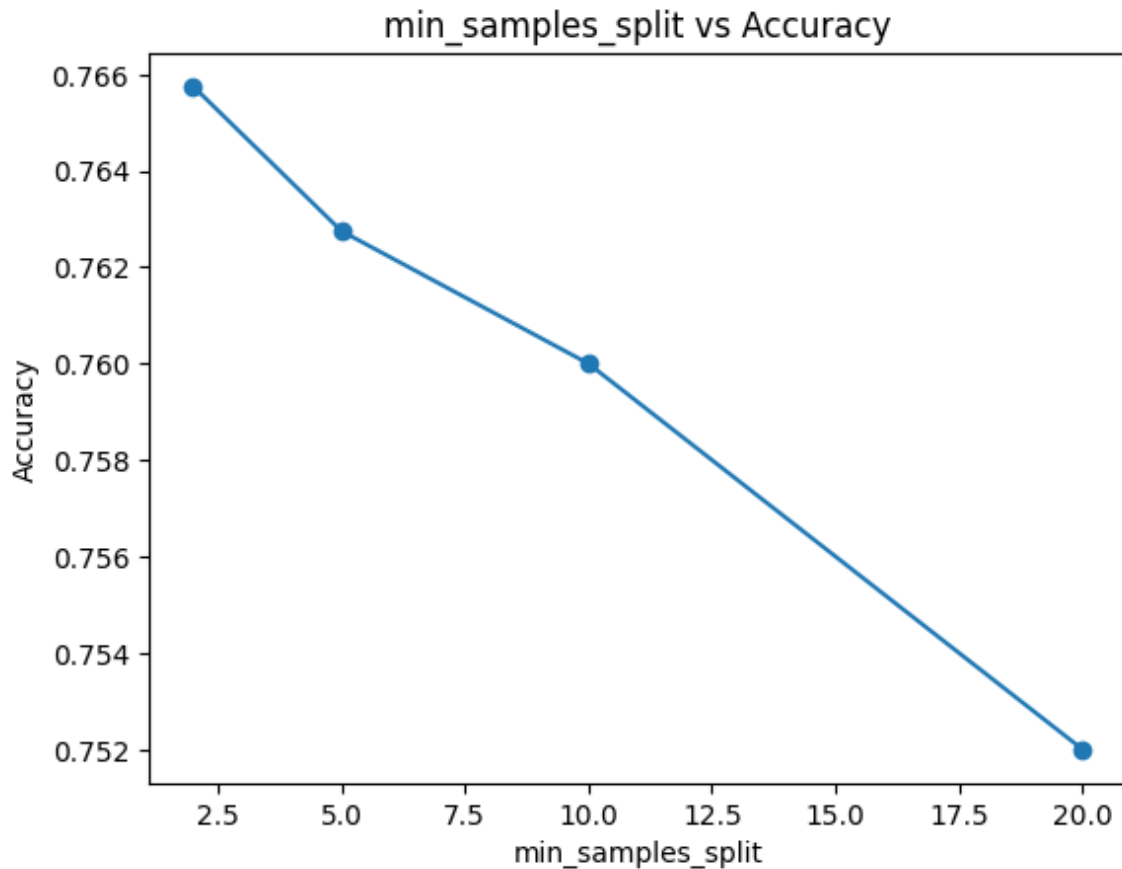


Figure 10. Random Forest Hyperparameter Search for Minimum Samples Split (`min_samples_split`).

The plot shows that lower values of minimum samples allow the model to create more complex trees, while higher values simplify the model and reduce overfitting. The results show that an intermediate value provides the best classification accuracy and generalization ability.

Overall, the hyperparameter tuning results show that the Random Forest model achieves optimal performance when properly balancing model complexity and generalization through `n_estimators`, `max_depth`, and `min_samples_split`.

6.0 Result and discussion

This section presents the experimental results of the proposed classification models and discusses their performance. The experiments were carried out in two stages. First, the three models, Logistic Regression, Decision Tree, and Random Forest, were trained and evaluated using the complete set of selected input features. Then, in Section 6.1, a feature selection experiment was conducted to examine how reducing the number of input features affects predictive performance, and to identify the smallest feature subset that maintains or improves accuracy.

The initial experiment used the full set of pre-flight features (month, day, day of week, airline, distance, origin airport, destination airport, scheduled departure, and departure time), with the target variable defined as whether a flight had an arrival delay greater than zero. The models were evaluated on the held-out 20% test set using accuracy, precision, recall, F1-score, ROC-AUC, MCC, Kappa, and Log Loss, consistent with the metrics reported in Section 5.3. The results are summarized in Table 1.

Table 10: Results of using the complete feature set

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC	MCC	Kappa	Log Loss
Logistic Regression	0.6670	0.5253	0.6231	0.5700	0.7196	0.3045	0.3015	0.6485
Decision Tree	0.7133	0.5898	0.6260	0.6073	0.6936	0.3822	0.3818	10.3355
Random Forest	0.7382	0.6670	0.5215	0.5853	0.7574	0.4047	0.3983	0.5719

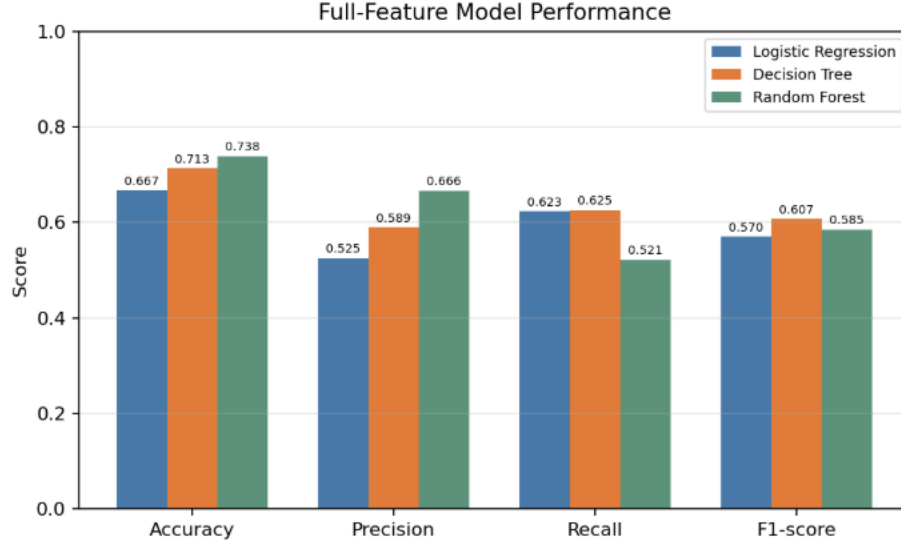


Figure 11: Performance comparison of the three models using the complete feature set.

As shown in Table 1, it can be seen that the performance of Random Forest algorithm is better than others as the highest accuracy achieved using this algorithm is 0.738, then comes Decision Tree with an accuracy of 0.713, and the lowest accuracy obtained by the Logistic Regression algorithm is 0.667. This is due to the fact that Random Forest is not vulnerable to overfitting. It is worth noting, however, that although Random Forest led on accuracy and precision, the Decision Tree obtained the highest F1-score (0.607), indicating a better balance between precision and recall for the delayed class. This trade-off is expected given the mild class imbalance (approximately 64% non-delayed and 36% delayed flights): a model can reach a high accuracy by favoring the majority (non-delayed) class while still missing a meaningful share of delayed flights. These full-feature results serve as the baseline against which the feature selection experiment in Section 6.1 is compared.

6.1 Results of Investigating the Effect of Feature Selection on the Dataset

Even though the full-feature models produced acceptable results, a feature selection experiment was carried out to determine whether a smaller subset of features could achieve comparable or better performance. Fewer input variables mean lower complexity and computational costs, higher comprehensibility, and can mitigate overfitting by eliminating irrelevant predictors. In order to ensure that the test is on an equal footing for all feature subsets, each one was tested with the exact same 80/20 split, preprocessing steps, and model parameters as in the full feature test described in Section 6.0 using accuracy as the metric of choice.

A recursive feature elimination procedure based on the correlation of each feature with the target variable was applied, following these steps:

1. Build the three classifiers using all V features (here $V = 9$).
2. Compute the correlation of each feature with the target variable and arrange the features in descending order of correlation strength.
3. Retain the best $V/2$ features and re-evaluate the models; repeat the halving process until only one feature remains.
4. Select, for each model, the feature subset that yields the best test-set performance and use it to build the final model.

It is important to note that this ranking is computed only over the retained pre-flight predictors. The high-correlation variables reported earlier in the correlation analysis of Section 5.1.2 (such as departure delay, airline delay, and late aircraft delay) were excluded beforehand because they represent post-flight information and would cause target leakage. The ranking of the remaining valid features by their absolute correlation with arrival delay is shown in Table 2. Among these predictors, departure time and scheduled departure were by far the most informative, while day of week and distance contributed almost nothing on their own.

Table 11: Features ranked by absolute correlation with arrival delay

Feature	Correlation with Arrival Delay
Departure Time	0.163
Scheduled Departure	0.129
Month	0.057
Destination Airport	0.041
Airline	0.036
Origin Airport	0.035
Day	0.011
Day of Week	0.001
Distance	0.001

Using this ranking, the models were re-evaluated on progressively smaller feature subsets: all 9 features, the best V/2 (4) features, the top 2 features, and finally the single top feature. The test-set accuracy for each subset is reported in Table 3 and illustrated in Figure 2.

Table 12: Test-set accuracy for different feature subsets

Feature subset	Logistic Regression	Decision Tree	Random Forest
All features (9)	0.667	0.713	0.738
Best V/2 features (4)	0.669	0.729	0.791
Top 2 features	0.677	0.779	0.810
Top 1 feature	0.569	0.620	0.568

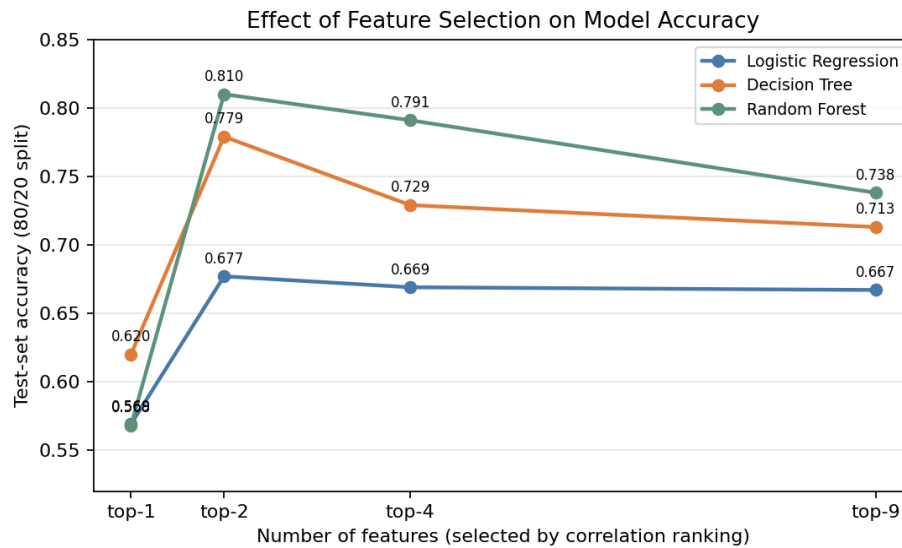


Figure 12: Effect of feature selection on model accuracy across feature subset sizes.

The results reveal a clear and useful pattern. For the tree-based models, reducing the feature set did not harm performance; it improved it substantially. Random Forest rose from 0.738 with all nine features to 0.791 with the best four, and reached its highest accuracy of 0.810 when restricted to only the two strongest features (departure time and scheduled departure). The Decision Tree showed the same trend, improving from 0.713 to 0.779 as weaker features were removed. This confirms that the lower-ranked features (day, day of week,

distance, and the airport identifiers) acted mainly as noise and encouraged the tree-based models to overfit. Logistic Regression remained essentially flat (around 0.67) across the larger subsets, which is consistent with its linear decision boundary being less sensitive to the removal of weak predictors. When the subset was reduced to a single feature, however, performance collapsed for every model (down to roughly 0.57 to 0.62), showing that no individual feature is sufficient and that at least the two strongest scheduling features are required to make useful predictions.

6.2 Discussion of Final Results

Bringing the experiments together, the final model was constructed by combining the two best choices identified in this study: the most effective feature subset from the feature selection experiment in Section 6.1 (the two highest-ranked features, departure time and scheduled departure) and the optimal Random Forest hyperparameters from the tuning experiment in Section 5.4 ($n_estimators = 400$, $max_depth = 20$, $min_samples_split = 10$, $class_weight = balanced$). This final configuration was trained and evaluated on the same 80/20 split as the previous experiments. Table 4 compares it against the full-feature baseline Random Forest from Section 6.0.

Table 13: Final tuned model versus the full-feature baseline

Configuration	Accuracy	Precision	Recall	F1-score	ROC-AUC
Baseline RF (9 features)	0.7382	0.6670	0.5215	0.5853	0.7574
Final RF (2 features, tuned)	0.8017	0.7600	0.6436	0.6970	0.8128

Combining feature selection with hyperparameter tuning produced the strongest overall model in this study. To isolate the effect of the features alone, the subset comparison in Section 6.1 used the baseline model settings; the final model here additionally applies the tuned Random Forest hyperparameters from Section 5.4. The final two-feature tuned Random Forest reached an accuracy of 0.8017 and an ROC-AUC of 0.8128, compared with 0.7382 and 0.7574 for the nine-feature baseline. Although its raw accuracy is essentially the same as the untuned two-feature model from Section 6.1 (0.810), the tuning rebalances the model toward the delayed class: recall rises from 0.5215 to 0.6436 and the F1-score from 0.5853 to 0.6970, which is the more meaningful improvement for this imbalanced problem. The confusion matrix of the final model is shown in Figure 3 and its ROC curve in Figure 4.

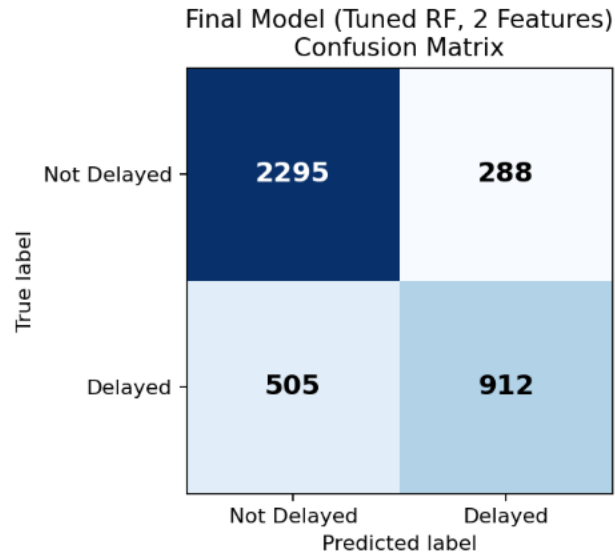


Figure 13: Confusion matrix of the final tuned Random Forest model.

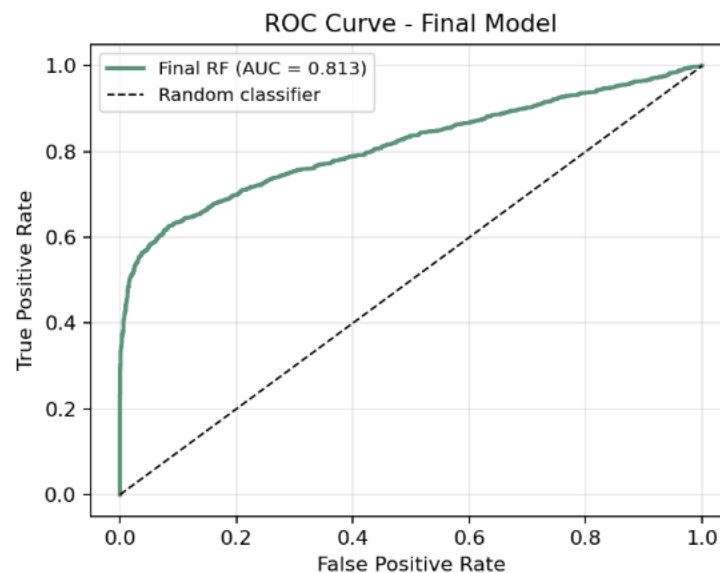


Figure 14: ROC curve of the final tuned Random Forest model.

The confusion matrix shows that the final model correctly classified 2295 non-delayed flights and 912 delayed flights, with 288 false positives and 505 false negatives. Relative to the baseline, the model recovers many delayed flights that were previously missed, which is the more costly type of error in an operational setting where unanticipated delays disrupt scheduling and passenger plans.

A simple what-if comparison highlights the practical value of this outcome. Reducing the input set from nine features to just two not only simplified the model but, once paired with

the tuned hyperparameters, actively improved every metric, which is an unusually favourable result for feature selection. If interpretability and data-collection cost are the priority, the two-feature model is ideal, since it relies on only the planned and actual departure times. If maximum predictive performance is the goal, this same configuration is also the best performer, so there is no trade-off to manage between simplicity and accuracy in this case.

The implication of these findings is that flight delay prediction in this dataset is driven overwhelmingly by scheduling-related information, specifically the planned and actual departure timing. Features such as distance, day of week, and the specific airports added little predictive value and, in the tree-based models, even introduced noise. For an airline operator who would use such a model, this is a highly beneficial result: a reliable delay indicator can be produced from just two readily available pre-flight attributes, which minimises the data collection and processing burden of deploying the model in practice.

It should be noted that the model was evaluated on a moderately imbalanced dataset, so accuracy alone does not tell the full story; the precision, recall, F1-score, and ROC-AUC reported above give a more complete picture of how well the delayed class is identified. Overall, the two-feature tuned Random Forest is selected as the final model for this study, as it offers the best balance of accuracy, simplicity, and practical applicability.

6.3 Further Discussions

This section presents a comparative discussion between the final results obtained in this study and previous research conducted using the same 2015 U.S. Flight Delays dataset. The comparison helps evaluate the effectiveness of the implemented machine learning models and determine the extent to which the proposed intelligent solution achieved competitive predictive performance.

The final results obtained from Logistic Regression, Decision Tree, and Random Forest classifiers were compared with the benchmark study conducted by AlBassam and AlShahrani (2025), which applied advanced resampling techniques on the same Kaggle flight delay dataset.

Table 14: Comparison Between Previous Studies and Current Models

Study	Accuracy	F1-Score	ROC-AUC	MCC
AlBassam & AlShahrani (2025) ★	0.90	0.90	0.87	0.73
Logistic Regression	0.6670	0.5700	0.7196	0.3045
Decision Tree	0.7133	0.6073	0.6936	0.3822
Random Forest	0.7382	0.5853	0.7574	0.4047

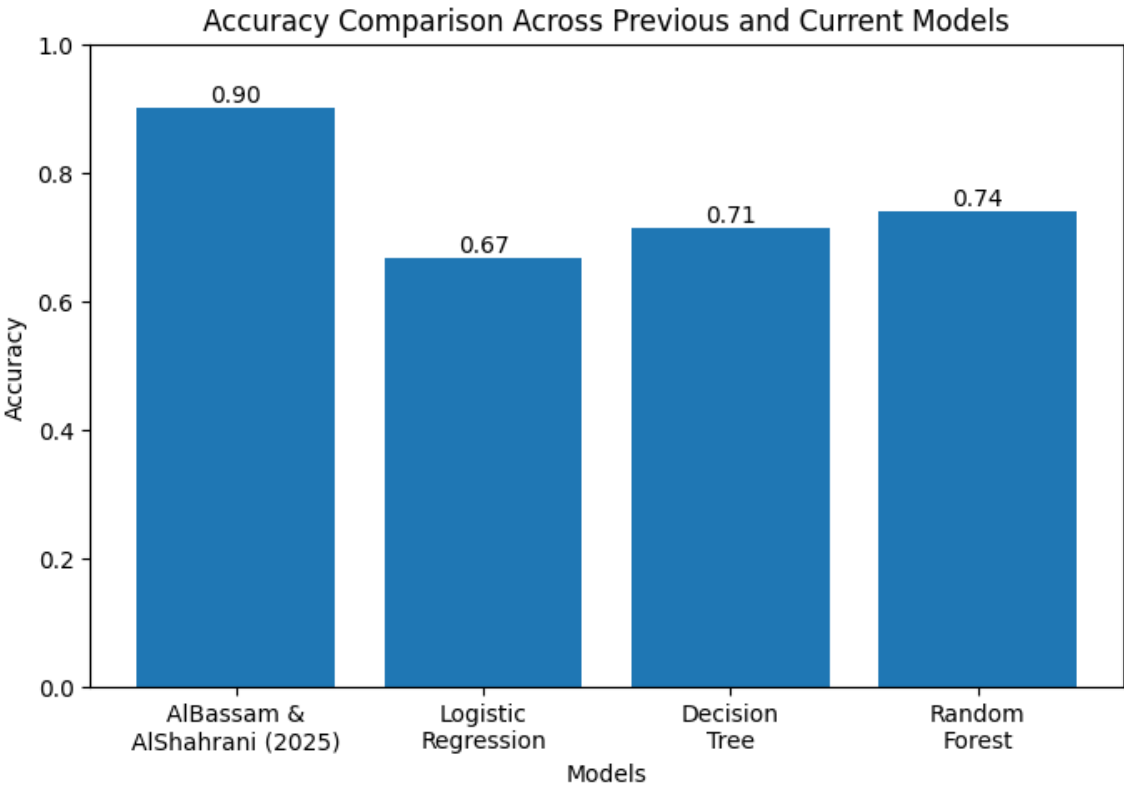


Figure 15: Accuracy Comparison Across Previous and Current Models

The comparison results indicate that the Random Forest classifier achieved the best overall performance among the implemented models in this project. The model achieved an accuracy of 73.82%, outperforming both Logistic Regression and Decision Tree classifiers. This result confirms the effectiveness of ensemble learning techniques in handling large and complex flight delay datasets.

However, the benchmark study by AlBassam and AlShahrani achieved superior predictive performance, reaching 90% accuracy and 90% F1-score. The higher performance reported in their work can be attributed mainly to the use of advanced class imbalance handling techniques such as Random Oversampling and SMOTE. In contrast, the current study focused primarily on hyperparameter optimization and class weighting without implementing advanced resampling techniques.

The ROC-AUC comparison further demonstrates that Random Forest provided better discrimination capability between delayed and non-delayed flights compared with the other implemented models. This confirms that ensemble learning methods are more capable of capturing non-linear relationships and hidden patterns within airline operational data.

Another important observation is that Decision Tree achieved the highest Recall and F1-score among the implemented models, indicating its ability to correctly identify delayed flights more effectively than Logistic Regression. Nevertheless, Random Forest maintained the best balance across all evaluation metrics, making it the most reliable model overall.

The findings of this study demonstrate that machine learning techniques can effectively support airline operational planning and scheduling optimization. Accurate flight delay prediction systems may help airlines improve resource allocation, reduce operational disruptions, optimize airport management, and enhance passenger satisfaction through earlier delay notifications and improved scheduling decisions.

Overall, although the proposed Random Forest model did not outperform the benchmark study, the achieved results remain competitive and demonstrate that hyperparameter optimization significantly improves model performance and generalization capability even without applying complex resampling pipelines.

6.4 Alignment with the requirements:

The intelligent solution implemented in this project aligns closely with the client's requirements because it applies machine learning techniques to solve a real-world aviation problem related to flight delay prediction. The client required the development of an intelligent predictive system capable of analyzing historical flight information and predicting whether flights would be delayed or on time.

To satisfy these requirements, three supervised machine learning classification models were implemented and compared, namely Logistic Regression, Decision Tree, and Random Forest. These models represent different machine learning approaches, including linear classification, rule-based learning, and ensemble learning techniques.

The implemented solution also fulfills the requirement of comparing multiple machine learning techniques using several evaluation metrics instead of relying only on prediction accuracy. The models were evaluated using Accuracy, Precision, Recall, F1-score, ROC-AUC, MCC, Kappa score, Log Loss, and Confusion Matrices to ensure reliable and comprehensive performance analysis.

Furthermore, the project includes all major stages required in intelligent system development, including data preprocessing, feature selection, model training, model testing, model evaluation, and hyperparameter optimization using GridSearchCV. This demonstrates that the implemented solution follows a complete machine learning workflow aligned with the project objectives and client requirements.

The final intelligent system can support airlines and airports in improving operational efficiency, optimizing scheduling decisions, reducing delay-related disruptions, and improving passenger satisfaction through early prediction of flight delays.

7.0 Conclusion and recommendation

This study investigated the application of machine learning techniques for flight delay prediction using the 2015 U.S. Flight Delays dataset obtained from the Kaggle platform. Three supervised classification models were implemented and evaluated, namely Logistic Regression, Decision Tree, and Random Forest.

The experimental results demonstrated that the Random Forest classifier achieved the best overall predictive performance with an accuracy of 73.82% and ROC-AUC of 75.74%, outperforming both Logistic Regression and Decision Tree models. The superior performance of Random Forest can be explained by its ensemble learning mechanism, which combines multiple decision trees to improve prediction capability and reduce overfitting.

The findings of this study confirm that machine learning techniques can effectively support flight delay prediction systems and contribute to improving airline operational planning, airport scheduling, resource allocation, and passenger experience. Accurate prediction systems may help airlines reduce operational costs, minimize scheduling disruptions, and provide better communication with passengers regarding flight status.

Despite the promising results obtained in this project, several limitations still exist. One of the major challenges was the class imbalance between delayed and non-delayed flights, which affected the predictive performance of some models. In addition, only a sampled subset of the original dataset was used to reduce computational complexity.

Therefore, future work may investigate advanced machine learning and deep learning approaches such as XGBoost, LightGBM, Gradient Boosting, and Artificial Neural Networks to further improve predictive performance. Future studies may also apply advanced class imbalance handling techniques such as SMOTE, ADASYN, and Random Oversampling. In addition, integrating real-time weather conditions, airport congestion information, and live operational flight data may further improve model accuracy and generalization capability.

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